

LAPORAN PRAKTIKUM 1
STRUKTUR SISTEM OPERASI



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1. Operating-System Service

1.1: Menjelajahi Layanan OS

- `uname -a` # *Lihat informasi sistem*
- `lsb_release -a`
- `ps aux | head -20` # *Lihat proses yang berjalan*
- `free -h` # *Lihat penggunaan memori*

```
alangan@LAPTOP-CV1QG677:~$ uname -a
Linux LAPTOP-CV1QG677 6.6.87.2-microsoft-standard-hs12 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 GNU/Linux
alangan@LAPTOP-CV1QG677:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:        24.04
Codename:       noble
alangan@LAPTOP-CV1QG677:~$ ps aux | head -20
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.1 21704 11988 ?        Ss   18:43   0:01 /sbin/init
root           2  0.0  0.0   3072  1792 ?        Ss   18:43   0:00 /init
root           7  0.0  0.0   3072  1792 ?        Ss   18:43   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          46  0.1  0.2  58432 16472 ?        Ss   18:43   0:00 /usr/lib/systemd/systemd-journald
root          93  0.1  0.0   25136  6272 ?        Ss   18:43   0:00 /usr/lib/systemd/systemd-udev
systemd+     173  0.1  0.1  21456 12672 ?        Ss   18:43   0:00 /usr/lib/systemd/systemd-resolved
systemd+     174  0.0  0.0   91024  7552 ?        Ssl  18:43   0:00 /usr/lib/systemd/systemd-timesyncd
root         186  0.0  0.0   4236  2432 ?        Ss   18:43   0:00 /usr/sbin/cron -f -P
message+    187  0.0  0.0   9624  4992 ?        Ss   18:43   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root        197  0.0  0.1  17960  8320 ?        Ss   18:43   0:00 /usr/lib/systemd/systemd-logind
root        199  0.0  0.1  1756096 12032 ?       Ssl  18:43   0:00 /usr/libexec/wsl-pro-service -vv
syslog      210  0.0  0.0  222508  5120 ?        Ssl  18:43   0:00 /usr/sbin/rsyslogd -n -iNONE
root       218  0.0  0.0   3160  1920 hvc0     S+   18:43   0:00 /sbin/agetty -o -p -- \u --noclean --keep-baud - 115200,38400,9600 vt220
root       221  0.0  0.0   3116  1792 tty1     S+   18:43   0:00 /sbin/agetty -o -p -- \u --noclean - linux
root       227  0.0  0.2 107012 22272 ?        Ssl  18:43   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root       324  0.0  0.0   6660  4608 pts/1    Ss   18:43   0:00 /bin/login -f
alangan+   371  0.1  0.1  20108 10752 ?        Ss   18:43   0:00 /usr/lib/systemd/systemd --user
alangan+   372  0.0  0.0   21156  3520 ?        S    18:43   0:00 (sd-pam)
alangan+   391  0.0  0.0   6056  4992 pts/1    S+   18:43   0:00 -bash
alangan@LAPTOP-CV1QG677:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.4Gi        524Mi        6.7Gi         3.5Mi        402Mi        6.9Gi
Swap:          2.0Gi          0B          2.0Gi
```

- `lscpu` # *Lihat informasi CPU*

```
alangan@LAPTOP-CV1QG677:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 48 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 16
On-line CPU(s) list: 0-15
Vendor ID: AuthenticAMD
Model name: AMD Ryzen 7 5700U with Radeon Graphics
CPU family: 23
Model: 100
Thread(s) per core: 2
Core(s) per socket: 8
Socket(s): 1
Stepping: 1
BogoMIPS: 3593.13
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep_good nopl tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes x save avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw topoext perfctr_core ssbd ibrs ibpb stibp vmcall fsgsbase bmi1 avx2 smep bmi2 rdtseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 clzero xsaveprtr arat npt nrrip_save tsc_sc ale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload umip rdpid
Virtualization features:
Virtualization: AMD-V
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d: 256 KiB (8 instances)
L1i: 256 KiB (8 instances)
L2: 4 MiB (8 instances)
L3: 4 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0-15
Vulnerabilities:
Gather data sampling: Not affected
Tlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Reg file data sampling: Not affected
Retbleed: Mitigation; untrained return thunk; SMT enabled with STIBP protection
Spec rstack overflow: Vulnerable; Safe RET, no microcode
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Retpolines; IBPB conditional; STIBP always-on; RSB filling; PBSRB-eIBRS Not affected; BHI Not affected
Srbds: Not affected
Tsx async abort: Not affected
```

- lsblk # Lihat perangkat yang terpasang

```
alanganteng@LAPTOP-CV1QG677:~$ lsblk
NAME
MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda  8:0    0 388.4M  1 disk
sdb  8:16   0   186M  1 disk
sdc  8:32   0    2G    0 disk [SWAP]
sdd  8:48   0    1T    0 disk /mnt/wslg/distro
```

1.2: Program C – Menggunakan Layanan OS

```
alanganteng@LAPTOP-CV1QG677:~$ gcc -o os_services alan_so.c
alanganteng@LAPTOP-CV1QG677:~$ ./os_services
=== Program Execution Service ===
Process ID (PID): 3297
Parent Process ID (PPID): 466

=== I/O Operations Service ===
File created and written successfully

=== File System Manipulation ===
File size: 23 bytes
File permissions: 644

=== Communications ===
Hostname: (null)

=== Error Detection ===
Error opening file: No such file or directory

=== Resource Allocation ===
Memory allocated successfully
alanganteng@LAPTOP-CV1QG677:~$
```

2. User and Operating-System Interface

2.1: Command Line Interface (CLI)

- pwd # Navigasi dasar
- ls -la
- cd /tmp
- mkdir test_dir
- cd test_dir
- ls -l # Pipelining

- ps aux | grep bash | wc -l
- echo "Hello World" > output.txt # *Redirection*
- cat output.txt
- cat output.txt >> output.txt
- cat < output.txt
- sleep 30 & # *Background processes*
- jobs
- fg %1

```

alanganteng@LAPTOP-CV1QG677:~$ pwd
/home/alanganteng
alanganteng@LAPTOP-CV1QG677:~$ ls -la
total 60
drwxr-x--- 4 alanganteng alanganteng 4096 Oct  8 19:06 .
drwxr-xr-x 3 root        root        4096 Oct  8 16:43 ..
-rw----- 1 alanganteng alanganteng   99 Oct  8 18:46 .bash_history
-rw-r--r-- 1 alanganteng alanganteng  220 Oct  8 16:43 .bash_logout
-rw-r--r-- 1 alanganteng alanganteng 3771 Oct  8 16:43 .bashrc
drwx----- 2 alanganteng alanganteng 4096 Oct  8 16:44 .cache
drwxr-xr-x 3 alanganteng alanganteng 4096 Oct  8 18:54 .local
-rw-rw-r-- 1 alanganteng alanganteng    0 Oct  8 16:44 .motd_shown
-rw-r--r-- 1 alanganteng alanganteng  807 Oct  8 16:43 .profile
-rw-r--r-- 1 alanganteng alanganteng    0 Oct  8 18:58 .sudo_as_admin_successful
-rw-r--r-- 1 alanganteng alanganteng 1314 Oct  8 18:57 alan_so.c
-rwxr-xr-x 1 alanganteng alanganteng 16480 Oct  8 19:05 os_services
-rw-r--r-- 1 alanganteng alanganteng   23 Oct  8 19:06 test_file.txt
alanganteng@LAPTOP-CV1QG677:~$ cd /tmp
alanganteng@LAPTOP-CV1QG677:/tmp$ mkdir test_dir
alanganteng@LAPTOP-CV1QG677:/tmp$ cd test_dir
alanganteng@LAPTOP-CV1QG677:/tmp/test_dir$ cd
alanganteng@LAPTOP-CV1QG677:~$ ls -l | grep "^d"
alanganteng@LAPTOP-CV1QG677:~$ ps aux | grep bash | wc -l
wc: invalid option -- 'l'
Try 'wc --help' for more information.
alanganteng@LAPTOP-CV1QG677:~$ ls -l | grep "^d"
alanganteng@LAPTOP-CV1QG677:~$ ps aux | grep bash | wc -l
wc: invalid option -- 'l'
Try 'wc --help' for more information.
alanganteng@LAPTOP-CV1QG677:~$ ps aux | grep bash | wc -l
3
alanganteng@LAPTOP-CV1QG677:~$ cd
alanganteng@LAPTOP-CV1QG677:~$ echo "Hello World" > output.txt
alanganteng@LAPTOP-CV1QG677:~$ cat output.txt
Hello World
alanganteng@LAPTOP-CV1QG677:~$ cat output.txt >> output.txt
cat: output.txt: input file is output file
alanganteng@LAPTOP-CV1QG677:~$ cat < output.txt
Hello World
alanganteng@LAPTOP-CV1QG677:~$ cd
alanganteng@LAPTOP-CV1QG677:~$ sleep 30 &
[1] 3334
alanganteng@LAPTOP-CV1QG677:~$ jobs
[1]+  Running                  sleep 30 &
alanganteng@LAPTOP-CV1QG677:~$ fg %1
sleep 30

```

2.2 Program C – Shell Sederhana

- Buat file simple_shell.c

```
alanganteng@LAPTOP-CV1QG677:~$ nano simple_shell.c
alanganteng@LAPTOP-CV1QG677:~$ gcc -o simple_shell simple_shell.c
alanganteng@LAPTOP-CV1QG677:~$ ./simple_shell
myshell> exit
alanganteng@LAPTOP-CV1QG677:~$
```

3. System Calls

3.1: Process Control System Calls

- Buat file process_syscalls.c

3.2: File Management System Calls

- Buat file file_syscalls.c

3.2: Information Maintenance System Calls

- Buat file info_syscalls.c

Compile Semua:

- gcc -o process_syscalls process_syscalls.c
- gcc -o file_syscalls file_syscalls.c
- gcc -o info_syscalls info_syscalls.c

- ./process_syscalls
- ./file_syscalls
- ./info_syscalls

```

alangenteng@LAPTOP-CV1QG677:~$ nano process_syscalls.c
alangenteng@LAPTOP-CV1QG677:~$ nano file_syscalls.c
alangenteng@LAPTOP-CV1QG677:~$ nano info_syscalls.c
alangenteng@LAPTOP-CV1QG677:~$ gcc -o process_syscalls process_syscalls.c
alangenteng@LAPTOP-CV1QG677:~$ gcc -o file_syscalls file_syscalls.c
alangenteng@LAPTOP-CV1QG677:~$ gcc -o info_syscalls info_syscalls.c
alangenteng@LAPTOP-CV1QG677:~$ ./process_syscalls
=== Process Control System Calls ===
Parent PID: 3369
Parent waiting for child (PID: 3370)...
Child PID: 3370
Child's Parent PID: 3369
Child executing 'ls' command...
total 116
-rw-r--r-- 1 alangenteng alangenteng 1314 Oct 8 18:57 alan_so.c
-rwxr-xr-x 1 alangenteng alangenteng 16352 Oct 8 19:44 file_syscalls
-rw-r--r-- 1 alangenteng alangenteng 967 Oct 8 19:42 file_syscalls.c
-rwxr-xr-x 1 alangenteng alangenteng 16384 Oct 8 19:44 info_syscalls
-rw-r--r-- 1 alangenteng alangenteng 713 Oct 8 19:43 info_syscalls.c
-rwxr-xr-x 1 alangenteng alangenteng 16480 Oct 8 19:05 os_services
-rw-r--r-- 1 alangenteng alangenteng 12 Oct 8 19:27 output.txt
-rwxr-xr-x 1 alangenteng alangenteng 16360 Oct 8 19:44 process_syscalls
-rw-r--r-- 1 alangenteng alangenteng 831 Oct 8 19:40 process_syscalls.c
-rwxr-xr-x 1 alangenteng alangenteng 16528 Oct 8 19:36 simple_shell
-rw-r--r-- 1 alangenteng alangenteng 898 Oct 8 19:35 simple_shell.c
-rw-r--r-- 1 alangenteng alangenteng 23 Oct 8 19:06 test_file.txt
Child exited with status: 0
alangenteng@LAPTOP-CV1QG677:~$ ./file_syscalls
=== File Management System Calls ===
File descriptor: 3
Bytes written: 25
File closed
Read from file: Hello from system calls!
alangenteng@LAPTOP-CV1QG677:~$ ./info_syscalls
=== Information Maintenance System Calls ===

Process ID: 3372
Parent Process ID: 466
User ID: 1000
Group ID: 1000
Current time: Wed Oct 8 19:45:20 2025
Clock ticks since boot: 1718333996
Sleeping for 2 seconds...
Awake!

```

4.1 System Service

4.1: Menggunakan system

- systemctl list-units --type=service # Lihat semua service

- systemctl status cron # Status service

```

alangenteng@LAPTOP-CV1QG677:~$ systemctl list-units --type=service
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
console-getty.service              loaded active running Console Getty
console-setup.service              loaded active exited Set console font and keymap
cron.service                        loaded active running Regular background program processing daemon
dbus.service                       loaded active running D-Bus System Message Bus
getty@tty1.service                 loaded active running Getty on tty1
keyboard-setup.service             loaded active exited Set the console keyboard layout
kmod-static-nodes.service          loaded active exited Create List of Static Device Nodes
polkit.service                     loaded active running Authorization Manager
rsyslog.service                   loaded active running System Logging Service
setvtrgb.service                   loaded active exited Set console scheme
snapd.seeded.service              loaded active exited Wait until snapd is fully seeded
systemd-binfmt.service             loaded active exited Set Up Additional Binary Formats
systemd-journal-flush.service      loaded active exited Flush Journal to Persistent Storage
systemd-journald.service           loaded active running Journal Service
systemd-logind.service             loaded active running User Login Management
systemd-modules-load.service        loaded active exited Load Kernel Modules
systemd-remount-fs.service          loaded active exited Remount Root and Kernel File Systems
systemd-resolved.service           loaded active running Network Name Resolution
systemd-sysctl.service             loaded active exited Apply Kernel Variables
systemd-timesyncd.service           loaded active running Network Time Synchronization
systemd-tmpfiles-setup-dev-early.service loaded active exited Create Static Device Nodes in /dev gracefully
systemd-tmpfiles-setup-dev.service loaded active exited Create Static Device Nodes in /dev
systemd-tmpfiles-setup.service     loaded active exited Create Volatile Files and Directories
systemd-udev-trigger.service        loaded active exited Coldplug All udev Devices
systemd-udevd.service              loaded active running Rule-based Manager for Device Events and Files
systemd-update-utmp.service         loaded active exited Record System Boot/Shutdown in UTMP
systemd-user-sessions.service       loaded active exited Permit User Sessions
unattended-upgrades.service         loaded active running Unattended Upgrades Shutdown
user-runtime-dir@1000.service       loaded active exited User Runtime Directory /run/user/1000
user@1000.service                  loaded active running User Manager for UID 1000
wsl-pro.service                    loaded active running Bridge to Ubuntu Pro agent on Windows

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
          ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
          SUB    → The low-level unit activation state, values depend on unit type.

31 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
alangenteng@LAPTOP-CV1QG677:~$ systemctl status cron
● cron.service - Regular background program processing daemon
   Loaded: loaded (/usr/lib/systemd/system/cron.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-10-08 18:43:54 WIB; 3h 15min ago
     Docs: man:cron(8)
    Main PID: 186 (cron)
      Tasks: 1 (limit: 9113)

```

- journalctl -u cron -n 20 # Lihat log service

- systemctl list-unit-files --type=service --state=enable # Lihat service yang berjalan saat boot

```

alangenteng@LAPTOP-CV1QG677:~$ journalctl -u cron -n 20
Oct 08 17:17:01 LAPTOP-CV1QG677 CRON[500]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 08 17:17:01 LAPTOP-CV1QG677 CRON[500]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
Oct 08 17:17:01 LAPTOP-CV1QG677 CRON[500]: pam_unix(cron:session): session closed for user root
-- Boot 4d0c8e96823243609190e8fbc6c14250 --
Oct 08 17:21:48 LAPTOP-CV1QG677 systemd[1]: Started cron.service - Regular background program processing daemon.
Oct 08 17:21:48 LAPTOP-CV1QG677 (cron)[186]: cron.service: Referenced but unset environment variable evaluates to an empty string: EXTRA_OPTS
Oct 08 17:21:48 LAPTOP-CV1QG677 cron[186]: (CRON) INFO (pidfile fd = 3)
Oct 08 17:21:48 LAPTOP-CV1QG677 cron[186]: (CRON) INFO (Running @reboot jobs)
-- Boot 34ead0b94f1545df91ba73d902a53782 --
Oct 08 18:43:54 LAPTOP-CV1QG677 systemd[1]: Started cron.service - Regular background program processing daemon.
Oct 08 18:43:54 LAPTOP-CV1QG677 (cron)[186]: cron.service: Referenced but unset environment variable evaluates to an empty string: EXTRA_OPTS
Oct 08 18:43:54 LAPTOP-CV1QG677 cron[186]: (CRON) INFO (pidfile fd = 3)
Oct 08 18:43:54 LAPTOP-CV1QG677 cron[186]: (CRON) INFO (Running @reboot jobs)
Oct 08 19:17:01 LAPTOP-CV1QG677 CRON[3306]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 08 19:17:01 LAPTOP-CV1QG677 CRON[3306]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
Oct 08 19:17:01 LAPTOP-CV1QG677 CRON[3306]: pam_unix(cron:session): session closed for user root
Oct 08 20:17:00 LAPTOP-CV1QG677 CRON[3470]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 08 20:17:00 LAPTOP-CV1QG677 CRON[3470]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
Oct 08 20:17:00 LAPTOP-CV1QG677 CRON[3470]: pam_unix(cron:session): session closed for user root
Oct 08 21:17:00 LAPTOP-CV1QG677 CRON[3793]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Oct 08 21:17:00 LAPTOP-CV1QG677 CRON[3793]: (root) CMD (cd / && run-parts --report /etc/cron.hourly)
Oct 08 21:17:00 LAPTOP-CV1QG677 CRON[3793]: pam_unix(cron:session): session closed for user root
alangenteng@LAPTOP-CV1QG677:~$ systemctl list-unit-files --type=service --state=enable
UNIT FILE                                STATE    PRESET
apparmor.service                         enabled enabled
appport.service                          enabled enabled
cloud-config.service                     enabled enabled
cloud-final.service                       enabled enabled
cloud-init-local.service                  enabled enabled
cloud-init.service                       enabled enabled
console-setup.service                    enabled enabled
cron.service                             enabled enabled
dmesg.service                            enabled enabled
e2scrub_reap.service                     enabled enabled
getty@.service                           enabled enabled
keyboard-setup.service                   enabled enabled
landscape-client.service                  enabled enabled
networkd-dispatcher.service               enabled enabled
rsyslog.service                          enabled enabled
setvtrgb.service                         enabled enabled
snapd.apparmor.service                   enabled enabled
snapd.autoimport.service                 enabled enabled
snapd.core-fixup.service                  enabled enabled
snapd.recovery-chosen-trigger.service     enabled enabled

```

4.2: Membuat Custom System Service

- Buat program sederhana my_daemon.c
- dan compile: gcc -o my_daemon my_daemon.c
- Buat service file /etc/systemd/system/my_daemon.service

Kelola service

- sudo systemctl daemon-reload
- sudo systemctl start my_daemon
- sudo systemctl status my_daemon

```
alangenteng@LAPTOP-CV1QG677:~$ nano my_daemon.c
alangenteng@LAPTOP-CV1QG677:~$ gcc -o my_daemon my_daemon.c
alangenteng@LAPTOP-CV1QG677:~$ nano sudo /etc/systemd/system/my_daemon.service
alangenteng@LAPTOP-CV1QG677:~$ sudo nano /etc/systemd/system/my_daemon.service
[sudo] password for alangenteng:
alangenteng@LAPTOP-CV1QG677:~$ sudo systemctl daemon-reload
alangenteng@LAPTOP-CV1QG677:~$ sudo systemctl start my_daemon
Job for my_daemon.service failed because the control process exited with error code.
See "systemctl status my_daemon.service" and "journalctl -xeu my_daemon.service" for details.
alangenteng@LAPTOP-CV1QG677:~$ sudo systemctl status my_daemon
* my_daemon.service - My Custom Daemon
   Loaded: loaded (/etc/systemd/system/my_daemon.service; disabled; preset: enabled)
   Active: failed (Result: exit-code) since Wed 2025-10-08 22:05:51 WIB; 28s ago
     Process: 4398 ExecStart=/path/to/my_daemon (code=exited, status=203/EXEC)
        CPU: 1ms
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
```

- sudo journalctl -u my_daemon
- sudo systemctl stop my_daemon

```
alangenteng@LAPTOP-CV1QG677:~$ sudo journalctl -u my_daemon
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 1.
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 2.
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:05:50 LAPTOP-CV1QG677 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 3.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 4.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: Starting my_daemon.service - My Custom Daemon...
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Control process exited, code=exited, status=203/EXEC
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Scheduled restart job, restart counter is at 5.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Start request repeated too quickly.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: my_daemon.service: Failed with result 'exit-code'.
Oct 08 22:05:51 LAPTOP-CV1QG677 systemd[1]: Failed to start my_daemon.service - My Custom Daemon.
alangenteng@LAPTOP-CV1QG677:~$ sudo systemctl stop my_daemon
```

5. Linkers and Loaders

5.1: Proses Liking

- Buat file math_ops.h
- math_ops.c
- main.c

Proses kompilasi bertahap

- gcc -E main.c -o main.i *# 1. Preprocessing*
- gcc -E math_ops.c -o math_ops.i
- gcc -S main.c -o main.s *# 2. Compilation (assembly)*
- gcc -S math_ops.c -o math_ops.s
- gcc -c main.c -o main.o *# 3. Assembly (object files)*
- gcc -c math_ops.c -o math_ops.o
- gcc main.o math_ops.o -o program *# 4. Linking*
- nm main.o *# Lihat symbols dalam object file*
- nm math_ops.o

```
alangenteng@LAPTOP-CV1QG677:~$ nano math_ops.h
alangenteng@LAPTOP-CV1QG677:~$ nano math_ops.c
alangenteng@LAPTOP-CV1QG677:~$ nano main.c
alangenteng@LAPTOP-CV1QG677:~$ gcc -E main.c -o main.i
alangenteng@LAPTOP-CV1QG677:~$ gcc -E math_ops.c -o math_ops.i
alangenteng@LAPTOP-CV1QG677:~$ gcc -S main.c -o main.s
alangenteng@LAPTOP-CV1QG677:~$ gcc -S math_ops.c -o math_ops.s
alangenteng@LAPTOP-CV1QG677:~$ gcc -c main.c -o main.o
alangenteng@LAPTOP-CV1QG677:~$ gcc -c math_ops.c -o math_ops.o
alangenteng@LAPTOP-CV1QG677:~$ gcc main.o math_ops.o -o program
alangenteng@LAPTOP-CV1QG677:~$ nm main.o
                 U add
0000000000000000 T main
                 U multiply
                 U printf
alangenteng@LAPTOP-CV1QG677:~$ nm math_ops.o
0000000000000000 T add
0000000000000018 T multiply
```

- nm program
- ldd program # *Lihat dependencies*

```

alangenteng@LAPTOP-CV1QG677:~$ nm program
000000000003dc8 d _DYNAMIC
000000000003fb8 d _GLOBAL_OFFSET_TABLE_
000000000002000 R _IO_stdin_used
w _ITM_deregisterTMCloneTable
w _ITM_registerTMCloneTable
000000000002148 r __FRAME_END__
00000000000201c r __GNU_EH_FRAME_HDR
000000000004010 D __TMC_END__
00000000000038c r __abi_tag
000000000004010 B __bss_start
w __cxa_finalize@GLIBC_2.2.5
000000000004000 D __data_start
0000000000001100 t __do_global_dtors_aux
000000000003dc0 d __do_global_dtors_aux_fini_array_entry
000000000004008 D __dso_handle
000000000003db8 d __frame_dummy_init_array_entry
w __gmon_start__
U __libc_start_main@GLIBC_2.34
000000000004010 D _edata
000000000004018 B _end
00000000000011d4 T _fini
0000000000001000 T _init
0000000000001060 T _start
00000000000011a2 T add
0000000000004010 b completed.0
000000000004000 W data_start
0000000000001090 t deregister_tm_clones
0000000000001140 t frame_dummy
0000000000001149 T main
00000000000011ba T multiply
U printf@GLIBC_2.2.5
00000000000010c0 t register_tm_clones
alangenteng@LAPTOP-CV1QG677:~$ ldd program
linux-vdso.so.1 (0x00007ffd221ba000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000073a380c00000)
/lib64/ld-linux-x86-64.so.2 (0x000073a380f5b000)

```

5.2: Static vs Dynamic Linking

- gcc main.c math_ops.c -o program_static -static # *Static linking*
- gcc main.c math_ops.c -o program_dynamic # *Dynamic linking (default)*
- ls -lh program_static program_dynamic # *Bandingkan ukuran*
- ldd program_static # *Lihat dependencies*
- ldd program_dynamic

```

alangenteng@LAPTOP-CV1QG677:~$ gcc main.c math_ops.c -o program_static -static
alangenteng@LAPTOP-CV1QG677:~$ gcc main.c math_ops.c -o program_dynamic
alangenteng@LAPTOP-CV1QG677:~$ ls -lh program_static program_dynamic
-rwxr-xr-x 1 alangenteng alangenteng 16K Oct  8 20:13 program_dynamic
-rwxr-xr-x 1 alangenteng alangenteng 767K Oct  8 20:12 program_static
alangenteng@LAPTOP-CV1QG677:~$ ldd program_static
not a dynamic executable
alangenteng@LAPTOP-CV1QG677:~$ ldd program_dynamic
linux-vdso.so.1 (0x00007ffd2a3d6000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x00007b76e8000000)
/lib64/ld-linux-x86-64.so.2 (0x00007b76e82ef000)

```

5.3: Membuat dan Menggunakan Shared Library

- gcc -c -fPIC math_ops.c -o math_ops.o *# Compile sebagai shared library*
- gcc -shared -o libmathops.so math_ops.o
- gcc main.c -L. -lmathops -o program_shared *# Compile program yang menggunakan library*
- LD_LIBRARY_PATH=. ./program_shared *# Jalankan dengan LD_LIBRARY_PATH*

```
alangenteng@LAPTOP-CV1QG677:~$ gcc -c -fPIC math_ops.c -o math_ops.o
alangenteng@LAPTOP-CV1QG677:~$ gcc -shared -o libmathops.so math_ops.o
alangenteng@LAPTOP-CV1QG677:~$ gcc main.c -L. -lmathops -o program_shared
alangenteng@LAPTOP-CV1QG677:~$ LD_LIBRARY_PATH=. ./program_shared
5 + 3 = 8
5 * 3 = 15
```

- readelf -d program_shared *# Lihat informasi loader*

```
alangenteng@LAPTOP-CV1QG677:~$ readelf -d program_shared

Dynamic section at offset 0x2da8 contains 28 entries:
   Tag               Type                             Name/Value
0x0000000000000001 (NEEDED)                     Shared library: [libmathops.so]
0x0000000000000001 (NEEDED)                     Shared library: [libc.so.6]
0x000000000000000c (INIT)                          0x1000
0x000000000000000d (FINI)                          0x11e4
0x0000000000000019 (INIT_ARRAY)                       0x3d98
0x000000000000001b (INIT_ARRAYSZ)                     8 (bytes)
0x000000000000001a (FINI_ARRAY)                       0x3da0
0x000000000000001c (FINI_ARRAYSZ)                     8 (bytes)
0x0000000006ffffff5 (GNU_HASH)                       0x3b0
0x0000000000000005 (STRTAB)                           0x4b0
0x0000000000000006 (SYMTAB)                           0x3d8
0x000000000000000a (STRSZ)                             170 (bytes)
0x000000000000000b (SYMENT)                           24 (bytes)
0x0000000000000015 (DEBUG)                             0x0
0x0000000000000003 (PLTGOT)                           0x3fa8
0x0000000000000002 (PLTRELSZ)                         72 (bytes)
0x0000000000000014 (PLTREL)                           RELA
0x0000000000000017 (JMPREL)                           0x660
0x0000000000000007 (RELA)                             0x5a0
0x0000000000000008 (RELASZ)                           192 (bytes)
0x0000000000000009 (RELAENT)                          24 (bytes)
0x000000000000001e (FLAGS)                            BIND_NOW
0x000000006ffffffb (FLAGS_1)                          Flags: NOW PIE
0x000000006ffffffe (VERNEED)                          0x570
0x000000006fffffff (VERNEEDNUM)                       1
0x000000006ffffff0 (VERSYM)                          0x55a
0x000000006ffffff9 (RELACOUNT)                       3
0x0000000000000000 (NULL)                             0x0
alangenteng@LAPTOP-CV1QG677:~$
```

6. Why Applications Are Operating-System Specific

6.1: System Calls Differences

- Buat file `os_specific.c`
- `file program` # *Lihat format executable*
- `readelf -h program` # *Lihat header ELF*

```
alanganteng@LAPTOP-CV1QG677:~$ nano os_specific.c
alanganteng@LAPTOP-CV1QG677:~$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0c5145e0b63b4709f23076fc2, for GNU/Linux 3.2.0, not stripped
alanganteng@LAPTOP-CV1QG677:~$ nano os_specific.c
alanganteng@LAPTOP-CV1QG677:~$ file program
program: ELF 64-bit LSB pie executable, x86-64, version 1 (SYSV), dynamically linked, interpreter /lib64/ld-linux-x86-64.so.2, BuildID[sha1]=0c5145e0b63b4709f23076fc2, for GNU/Linux 3.2.0, not stripped
alanganteng@LAPTOP-CV1QG677:~$ readelf -h program
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00
  Class:                   ELF64
  Data:                     2's complement, little endian
  Version:                   1 (current)
  OS/ABI:                    UNIX - System V
  ABI Version:               0
  Type:                      DYN (Position-Independent Executable file)
  Machine:                   Advanced Micro Devices X86-64
  Version:                   0x1
  Entry point address:       0x1060
  Start of program headers:  64 (bytes into file)
  Start of section headers: 14072 (bytes into file)
  Flags:                      0x0
  Size of this header:       64 (bytes)
  Size of program headers:   56 (bytes)
  Number of program headers: 13
  Size of section headers:   64 (bytes)
  Number of section headers: 31
  Section header string table index: 30
```


- readelf -S program # Lihat section headers

```
alangenteng@LAPTOP-CV1QG677:~$ readelf -S program
There are 31 section headers, starting at offset 0x36f8:
```

Section Headers:

[Nr]	Name	Type	Address	Offset
	Size	EntSize	Flags Link Info Align	
[0]	0000000000000000	NULL	0000000000000000	00000000
[1]	.interp	PROGBITS	0000000000000318	00000318
[2]	.note.gnu.pr[...]	NOTE	0000000000000338	00000338
[3]	.note.gnu.bu[...]	NOTE	0000000000000368	00000368
[4]	.note.ABI-tag	NOTE	000000000000038c	0000038c
[5]	.gnu.hash	GNU_HASH	00000000000003b0	000003b0
[6]	.dynsym	DYNSYM	00000000000003d8	000003d8
[7]	.dynstr	STRTAB	0000000000000480	00000480
[8]	.gnu.version	VERSYM	0000000000000510	00000510
[9]	.gnu.version_r	VERNEED	0000000000000520	00000520
[10]	.rela.dyn	RELA	0000000000000550	00000550
[11]	.rela.plt	RELA	0000000000000610	00000610
[12]	.init	PROGBITS	0000000000001000	00001000
[13]	.plt	PROGBITS	0000000000001020	00001020
[14]	.plt.got	PROGBITS	0000000000001040	00001040
[15]	.plt.sec	PROGBITS	0000000000001050	00001050

-

- readelf -l program # Lihat program headers

```
alangenteng@LAPTOP-CV1QG677:~$ readelf -l program

Elf file type is DYN (Position-Independent Executable file)
Entry point 0x1060
There are 13 program headers, starting at offset 64

Program Headers:
  Type           Offset             VirtAddr           PhysAddr
     FileSiz      MemSiz          Flags             Align
PHDR           0x0000000000000040 0x0000000000000040 0x0000000000000040
     0x00000000000002d8 0x00000000000002d8 R               0x8
INTERP         0x0000000000000318 0x0000000000000318 0x0000000000000318
     0x000000000000001c 0x000000000000001c R               0x1
      [Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]
LOAD           0x0000000000000000 0x0000000000000000 0x0000000000000000
     0x0000000000000628 0x0000000000000628 R               0x1000
LOAD           0x0000000000000100 0x0000000000000100 0x0000000000000100
     0x00000000000001e1 0x00000000000001e1 R E             0x1000
LOAD           0x0000000000000200 0x0000000000000200 0x0000000000000200
     0x000000000000014c 0x000000000000014c R               0x1000
LOAD           0x00000000000002db8 0x00000000000003db8 0x00000000000003db8
     0x0000000000000258 0x0000000000000260 RW             0x1000
DYNAMIC         0x00000000000002dc8 0x00000000000003dc8 0x00000000000003dc8
     0x00000000000001f0 0x00000000000001f0 RW               0x8
NOTE           0x0000000000000338 0x0000000000000338 0x0000000000000338
     0x0000000000000030 0x0000000000000030 R               0x8
NOTE           0x0000000000000368 0x0000000000000368 0x0000000000000368
     0x0000000000000044 0x0000000000000044 R               0x4
GNU_PROPERTY    0x0000000000000338 0x0000000000000338 0x0000000000000338
     0x0000000000000030 0x0000000000000030 R               0x8
GNU_EH_FRAME    0x0000000000000201c 0x0000000000000201c 0x0000000000000201c
     0x0000000000000044 0x0000000000000044 R               0x4
GNU_STACK       0x0000000000000000 0x0000000000000000 0x0000000000000000
     0x0000000000000000 0x0000000000000000 RW              0x10
GNU_RELRO       0x00000000000002db8 0x00000000000003db8 0x00000000000003db8
     0x0000000000000248 0x0000000000000248 R               0x1
```

- objdump -d program | head -50 # Disassemble

```
alangenteng@LAPTOP-CV1QG677:~$ objdump -d program | head -50

program:          file format elf64-x86-64

Disassembly of section .init:

0000000000000100 <.init>:
1000:    f3 0f 1e fa                endbr64
1004:    48 83 ec 08                sub    $0x8,%rsp
1008:    48 8b 05 d9 2f 00 00      mov    0x2fd9(%rip),%rax        # 3fe8 <__gmon_start__@Base>
100f:    48 85 c0                  test   %rax,%rax
1012:    74 02                    je     1016 <.init+0x16>
1014:    ff d0                    call   *%rax
1016:    48 83 c4 08                add    $0x8,%rsp
101a:    c3                      ret

Disassembly of section .plt:

00000000000001020 <.plt>:
1020:    ff 35 9a 2f 00 00      push   0x2f9a(%rip)            # 3fc0 <_GLOBAL_OFFSET_TABLE_+0x8>
1026:    ff 25 9c 2f 00 00      jmp    *0x2f9c(%rip)          # 3fc8 <_GLOBAL_OFFSET_TABLE_+0x10>
102c:    0f 1f 40 00              nopl   0x0(%rax)
1030:    f3 0f 1e fa                endbr64
1034:    68 00 00 00 00          push   $0x0
1039:    e9 e2 ff ff              jmp    1020 <.init+0x20>
103e:    66 90                    xchg   %ax,%ax

Disassembly of section .plt.got:

00000000000001040 <__cxa_finalize@plt>:
1040:    f3 0f 1e fa                endbr64
1044:    ff 25 ae 2f 00 00      jmp    *0x2fae(%rip)          # 3ff8 <__cxa_finalize@GLIBC_2.2.5>
104a:    66 0f 1f 44 00 00      nopw   0x0(%rax,%rax,1)

Disassembly of section .plt.sec:

00000000000001050 <printf@plt>:
1050:    f3 0f 1e fa                endbr64
1054:    ff 25 76 2f 00 00      jmp    *0x2f76(%rip)          # 3fd0 <printf@GLIBC_2.2.5>
```


7. OS Design and Implementaion

7.1: Kernel Module Sederhana

- Buat file hello_module.c

Compile and Load

- make
- sudo insmod hello_module.ko
- lsmod | grep hello
- dmesg | tail
- sudo rmmod hello_module
- dmesg | tail

7.2: Monitoring Kernel

- uname -r *# Lihat informasi kernel*
- cat /proc/version

```
alantang@LAPTOP-CV1QG677:~$ nano hello_module.c
alantang@LAPTOP-CV1QG677:~$ make
Command 'make' not found, but can be installed with:
sudo apt install make          # version 4.3-4.1build1, or
sudo apt install make-guile    # version 4.3-4.1build1
alantang@LAPTOP-CV1QG677:~$ sudo insmod hello_module.ko
[sudo] password for alantang:
insmod: ERROR: could not load module hello_module.ko: No such file or directory
alantang@LAPTOP-CV1QG677:~$ lsmod | grep hello
alantang@LAPTOP-CV1QG677:~$ dmesg | tail
[ 16.746510] WSL (274) ERROR: CheckConnection: getaddrinfo() failed: -3
[ 48.894407] hv_balloon: Max. dynamic memory size: 7852 MB
[ 78.643494] WSL (274) ERROR: CheckConnection: connect() failed: 101
[ 593.740207] mini_init (258): drop_caches: 1
[ 883.832108] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 966.962961] systemd-journald[46]: Time jumped backwards, rotating.
[ 1067.395301] mini_init (258): drop_caches: 1
[ 1184.680725] mini_init (258): drop_caches: 1
[ 1272.732510] mini_init (258): drop_caches: 1
[ 6738.229054] systemd-journald[46]: Time jumped backwards, rotating.
alantang@LAPTOP-CV1QG677:~$ sudo rmmod hello_module
rmmod: ERROR: Module hello_module is not currently loaded
alantang@LAPTOP-CV1QG677:~$ dmesg | tail
[ 16.746510] WSL (274) ERROR: CheckConnection: getaddrinfo() failed: -3
[ 48.894407] hv_balloon: Max. dynamic memory size: 7852 MB
[ 78.643494] WSL (274) ERROR: CheckConnection: connect() failed: 101
[ 593.740207] mini_init (258): drop_caches: 1
[ 883.832108] TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.
[ 966.962961] systemd-journald[46]: Time jumped backwards, rotating.
[ 1067.395301] mini_init (258): drop_caches: 1
[ 1184.680725] mini_init (258): drop_caches: 1
[ 1272.732510] mini_init (258): drop_caches: 1
[ 6738.229054] systemd-journald[46]: Time jumped backwards, rotating.
alantang@LAPTOP-CV1QG677:~$ uname -r
6.6.87.2-microsoft-standard-WSL2
alantang@LAPTOP-CV1QG677:~$ cat /proc/version
Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46
```

- sysctl -a | head -20 *# Lihat parameter kernel*
- dmesg | tail -30 *# Lihat kernel ring buffer*

```

alangenteng@LAPTOP-CV1QG677:~$ sysctl -a | head -20
sysctl: permission denied on key 'kernel.cad_pid'
abi.vsyscall32 = 1
debug.exception-trace = 1
debug.kprobes-optimization = 1
dev.cdrom.autoclose = 1
dev.cdrom.autoeject = 0
dev.cdrom.check_media = 0
dev.cdrom.debug = 0
dev.cdrom.info = CD-ROM information, Id: cdrom.c 3.20 2003/12/17
dev.cdrom.info =
dev.cdrom.info = drive name:
dev.cdrom.info = drive speed:
dev.cdrom.info = drive # of slots:
dev.cdrom.info = Can close tray:
dev.cdrom.info = Can open tray:
dev.cdrom.info = Can lock tray:
dev.cdrom.info = Can change speed:
dev.cdrom.info = Can select disk:
dev.cdrom.info = Can read multisession:
dev.cdrom.info = Can read WCD:
dev.cdrom.info = Reports media changed:
alangenteng@LAPTOP-CV1QG677:~$ dmesg | tail -30
[ 3.348337] pulseaudio[385]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
[ 3.604789] misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
[ 3.614855] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.616588] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.618294] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.620086] misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
[ 3.708742] Failed to connect to bus: No such file or directory
[ 4.040660] systemd-journald[46]: Collecting audit messages is disabled.
[ 4.249020] systemd-journald[46]: Received client request to flush runtime journal.
[ 4.250823] systemd-journald[46]: File /var/log/journal/263dfe39f1b741789eee08dfb5beed78/system-journal corrupted or uncleanly shut down, renaming and replacing.
[ 4.758732] ACPI: battery: Slot [BAT1] (battery present)
[ 4.794222] ACPI: AC: AC Adaptor [AC1] (off-line)
[ 5.313880] kvm_amd: TSC scaling supported
[ 5.313895] kvm_amd: Nested Virtualization enabled
[ 5.313898] kvm_amd: Nested Paging enabled
[ 5.313908] kvm_amd: kvm_amd: Hyper-V enlightened NPT TLB flush enabled
[ 5.313911] kvm_amd: kvm_amd: Hyper-V Direct TLB flush enabled
[ 5.313941] kvm_amd: Virtual VMLOAD VMSAVE supported
[ 8.918356] systemd-journald[46]: /var/log/journal/263dfe39f1b741789eee08dfb5beed78/user-1000-journal: Journal file uses a different sequence number ID, n

```

- lsmod # Lihat kernel modules
- modinfo ext4 # Informasi modul tertentu

```

alangenteng@LAPTOP-CV1QG677:~$ lsmod
Module                  Size  Used by
tls                     98304  0
intel_rapl_msr         16384  0
intel_rapl_common      32768  1 intel_rapl_msr
kvm_amd                126976  0
kvm                    970752  1 kvm_amd
irqbypass              12288  1 kvm
crc32c_intel           16384  0
ac                     16384  0
battery                20480  0
sch_fq_codel           16384  1
configfs               53248  1
autofs4                45056  0
br_netfilter           28672  0
bridge                 282624  1 br_netfilter
stp                    12288  1 bridge
llc                    12288  2 bridge,stp
ip_tables              28672  0
tun                    53248  0
alangenteng@LAPTOP-CV1QG677:~$ modinfo ext4
name: ext4
filename: (builtin)
softdep: pre: crc32c
license: GPL
file: fs/ext4/ext4
description: Fourth Extended Filesystem
author: Remy Card, Stephen Tweedie, Andrew Morton, Andreas Dilger, Theodore Ts'o and others
alias: fs-ext4
alias: ext3
alias: fs-ext3

```

8. Operating-System Structure

8.1: Eksplorasi Struktur Layering

- Lspci # Layer 1: Hardware
- Lsusb
- cat /proc/cpuinfo

```
alanganteng@LAPTOP-CV1Q6677:~$ lspci
Command 'lspci' not found, but can be installed with:
sudo apt install pciutils
alanganteng@LAPTOP-CV1Q6677:~$ lsusb
Command 'lsusb' not found, but can be installed with:
sudo apt install usbutils
alanganteng@LAPTOP-CV1Q6677:~$ cat /proc/cpuinfo
processor       : 0
vendor_id      : AuthenticAMD
cpu family     : 23
model          : 104
model name     : AMD Ryzen 7 5700U with Radeon Graphics
stepping      : 1
microcode     : 0xffffffff
cpu MHz        : 1796.568
cache size     : 512 KB
physical id    : 0
siblings       : 16
core id        : 0
cpu cores      : 8
apicid         : 0
initial apicid : 0
cpu            : yes
cpu exception  : yes
cpuid level    : 13
wp             : yes
flags           : fpu vme de pse tsc msr pae mce cx8 sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp lm constant
tsc rep_good nopl tsc_reliable nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor
vfm lm cmov_legacy sum cmov_legacy_ahb sse4a misalignsse 3dnowprefetch osvw topoext perfctr_core sbbd ibrs ibpb stibp vmcall fsgsbase bmi1 avx2 smap bmi2 rdseed adx snap clf
ushopt clwb sha_ni xsaveopt xsavec xgetbv1 clzero xsaveerptr arat npt nrip_save tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pfthreshold v_vmsave_vmload umip
rdpid
bugs           : sysret_ss_attrs null_seg spectre_v1 spectre_v2 spec_store_bypass retbleed smt_rsb srso ibpb_no_ret
bogomips       : 3593.13
tlb size       : 3072 4K pages
clflush size    : 64
cache alignment : 64
address sizes   : 48 bits physical, 48 bits virtual
power management:

processor       : 1
vendor_id      : AuthenticAMD
cpu family     : 23
model          : 104
```

- uname -a # Layer 2: Kernel
- cat /proc/sys/kernel/ostype
- strace ls # Layer 3: System Calls
- ps aux # Layer 4: System Programs
- top

```
alanganteng@LAPTOP-CV1Q6677:~$ uname -a
Linux LAPTOP-CV1Q6677 6.6.87.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
alanganteng@LAPTOP-CV1Q6677:~$ cat /proc/sys/kernel/ostype
Linux
alanganteng@LAPTOP-CV1Q6677:~$ strace ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
alanganteng@LAPTOP-CV1Q6677:~$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1   0.0  0.1 21892 12928 ?        Ss   18:41   0:03 /usr/lib/systemd/systemd --system --deserialize=54
root           2   0.0  0.0   3072 1792 ?        Sl   18:41   0:00 /init
root           7   0.0  0.0   3072 1792 ?        Sl   18:41   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root          46   0.0  0.2 66828 18620 ?        Ss   18:41   0:01 /usr/lib/systemd/systemd-journald
root           93   0.0  0.0  25140  6272 ?        Ss   18:41   0:00 /usr/lib/systemd/systemd-udevd
systemd+      173   0.0  0.1 21456 12672 ?        Ss   18:41   0:00 /usr/lib/systemd/systemd-resolved
systemd+     174   0.0  0.0 91824 7552 ?         Ssl  18:41   0:00 /usr/lib/systemd/systemd-timesyncd
root         186   0.0  0.0  4236 2432 ?        Ss   18:41   0:00 /usr/sbin/cron -f -P
message+     187   0.0  0.0  9856 5248 ?        Ss   18:41   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root         197   0.0  0.1 17960 8320 ?        Ss   18:41   0:00 /usr/lib/systemd/systemd-logind
root         199   0.0  0.1 1756096 12672 ?       Ssl  18:41   0:00 /usr/libexec/wsl-pro-service -vv
root        218   0.0  0.0   3160 1920 hvcs0 Ss+  18:41   0:00 /sbin/agetty -o -p -- \u --noclean --keep-baud - 115200,38400,9600 vt220
root        221   0.0  0.0   3116 1792 tty1     Ss+  18:41   0:00 /sbin/agetty -o -p -- \u --noclean - linux
root        227   0.0  0.2 107012 22272 ?       Ssl  18:41   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
root        324   0.0  0.0   6660 4608 pts/1    Ss   18:41   0:00 /bin/login -f
alanganteng+ 371   0.0  0.1 20188 10752 ?        Ss   18:41   0:00 /usr/lib/systemd/systemd --user
alanganteng+ 372   0.0  0.0  21156 3520 ?         S    18:41   0:00 (sd-pam)
alanganteng+ 391   0.0  0.0   6056 4992 pts/1    S+   18:41   0:00 -bash
root         463   0.0  0.0   3076 904 ?         Ss   18:43   0:00 /init
root         464   0.0  0.0   3092 1164 ?         S    18:43   0:01 /init
alanganteng+ 466   0.0  0.0   6072 5248 pts/0    Ss   18:43   0:00 -bash
polkitd      852   0.0  0.0 308164 7680 ?        Ssl  18:58   0:00 /usr/lib/polkit-1/polkitd --no-debug
syslog       2739   0.0  0.0 222508 5248 ?        Ssl  19:00   0:00 /usr/sbin/rsyslogd -n -iNONE
alanganteng+ 3633 100.0  0.0  8280 4224 pts/0    R+   20:52   0:00 ps aux

alanganteng@LAPTOP-CV1Q6677:~$ top
top - 20:52:39 up 2:11, 1 user, load average: 0.00, 0.00, 0.00
tasks: 26 total, 1 running, 25 sleeping, 0 stopped, 0 zombie
  Cpu(s):  0.0 us,  0.1 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
  MiB Mem : 7684.2 total, 6917.4 free,  585.4 used, 264.1 buff/cache
  MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 7018.8 avail Mem

    1 root        20   0 21892 12928 9472 S   0.0  0.2  0:03.91 systemd
    2 root        20   0  3072 1792 1664 S   0.0  0.0  0:00.03 init-systemd(Ub
```

- which bash # Layer 5: User Applications
- which gcc

8.2: Microkernel vs Monolithic

- ls -lh /boot/vmlinuz-\$(uname -r) # Lihat ukuran kernel (Monolithic)
- cat /boot/config-\$(uname -r) | grep "=y" | wc -l # Built-in
- cat /boot/config-\$(uname -r) | grep "=m" | wc -l # Modules
- lsmod | wc -l # Lihat loaded modules

8.3: Program – Virtual File System

- buat file procfs_read.c

Compile dan Jalankan

- gcc -o procfs_read procfs_read.c
- ./procfs_read

```
alanganteng@LAPTOP-CV1Q6677:~$ which bash
/usr/bin/bash
alanganteng@LAPTOP-CV1Q6677:~$ which gcc
/usr/bin/gcc
alanganteng@LAPTOP-CV1Q6677:~$ ls -lh /boot/vmlinuz-$(uname -r)
ls: cannot access '/boot/vmlinuz-6.6.87.2-microsoft-standard-WSL2': No such file or directory
alanganteng@LAPTOP-CV1Q6677:~$ cat /boot/config-$(uname -r) | grep "=y" | wc -l # Built-in
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
alanganteng@LAPTOP-CV1Q6677:~$ cat /boot/config-$(uname -r) | grep "=y" | wc -l
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
alanganteng@LAPTOP-CV1Q6677:~$ cat /boot/config-$(uname -r) | grep "=m" | wc -l # Modules
cat: /boot/config-6.6.87.2-microsoft-standard-WSL2: No such file or directory
0
alanganteng@LAPTOP-CV1Q6677:~$ lsmod | wc -l
19
alanganteng@LAPTOP-CV1Q6677:~$ nano procfs_read.c
alanganteng@LAPTOP-CV1Q6677:~$ gcc -o procfs_read procfs_read.c
procfs_read.c: In function 'main':
procfs_read.c:31:60: warning: implicit declaration of function 'getpid' [-Wimplicit-function-declaration]
   31 |     snprintf(proc_path, sizeof(proc_path), "/proc/%d/status", getpid());
      |                                         ~~~~~
alanganteng@LAPTOP-CV1Q6677:~$ ./procfs_read
=== CPU Information ===
processor       : 0
vendor_id      : AuthenticAMD
cpu family     : 23
model          : 104
model name     : AMD Ryzen 7 5700U with Radeon Graphics
stepping       : 1
microcode      : 0xffffffff
cpu MHz        : 1796.568
cache size     : 512 KB
physical id    : 0

=== Memory Information ===
MemTotal:      7786684 kB
MemFree:       7086004 kB
MemAvailable:  7190544 kB
Buffers:       7368 kB
Cached:        215716 kB

=== Process Information ===
Name:  procfs_read
```

9. Building and Booting an Operating System

9.1: Analisis Boot Process

- dmesg | less # Lihat boot messages
- systemd-analyze # Lihat systemd boot analysis

- systemd-analyze blame

```
alantang@LAPTOP-CV1Q6677:~$ dmesg | less
[1]+  Stopped                  dmesg | less
alantang@LAPTOP-CV1Q6677:~$ systemd-analyze
Startup finished in 4.065s (userspace)
graphical.target reached after 4.002s in userspace.
alantang@LAPTOP-CV1Q6677:~$ systemd-analyze blame
8.706s motd-news.service
1.278s landscape-client.service
672ms dev-sdd.device
618ms snapd.seeded.service
454ms snapd.service
392ms user@1000.service
382ms systemd-udev-trigger.service
355ms wsl-pro.service
320ms systemd-resolved.service
232ms snapd.socket
198ms systemd-timesyncd.service
169ms systemd-tmpfiles-clean.service
165ms systemd-journald.service
158ms polkit.service
156ms systemd-journal-flush.service
152ms keyboard-setup.service
138ms systemd-udev.service
133ms rsyslog.service
120ms systemd-logind.service
106ms dev-hugepages.mount
104ms dev-mqueue.mount
100ms sys-kernel-debug.mount
96ms sys-kernel-tracing.mount
82ms systemd-tmpfiles-setup.service
81ms dbus.service
73ms e2scrub_reap.service
73ms kmod-static-nodes.service
72ms modprobe@configfs.service
71ms systemd-tmpfiles-setup-dev-early.service
70ms modprobe@dm_mod.service
69ms modprobe@drm.service
68ms modprobe@efi_pstore.service
68ms systemd-sysctl.service
63ms modprobe@fuse.service
61ms modprobe@loop.service
54ms user-runtime-dir@1000.service
53ms systemd-modules-load.service
48ms systemd-remount-fs.service
```

- systemd-analyze critical-chain
- cat /boot/grub/grub.cfg | head -50 # Lihat GRUB configuration
- lsinitramfs /boot/initrd.img-\$(uname -r) | head -20 # Lihat initramfs

9.2: Kernel Parameters

- cat /proc/cmdline # Lihat kernel parameters saat boot
- sudo nano /etc/default/grub # Permanent: Edit /etc/default/grub

```
alantang@LAPTOP-CV1Q6677:~$ systemd-analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

graphical.target @4.002s
├─multi-user.target @4.000s
│   └─snapd.seeded.service @2.206s +618ms
│       └─basic.target @2.161s
│           └─sockets.target @2.160s
│               └─snapd.socket @1.926s +232ms
│                   └─sysinit.target @1.903s
│                       └─systemd-resolved.service @1.581s +320ms
│                           └─systemd-tmpfiles-setup.service @982ms +82ms
│                               └─systemd-journal-flush.service @790ms +156ms
│                                   └─systemd-journald.service @533ms +165ms
│                                       └─systemd-journald.socket @496ms
│                                           └─.mount @445ms
│                                               └─.slice @443ms
alantang@LAPTOP-CV1Q6677:~$ cat /boot/grub/grub.cfg | head -50
cat: /boot/grub/grub.cfg: No such file or directory
alantang@LAPTOP-CV1Q6677:~$ lsinitramfs /boot/initrd.img-$(uname -r) | head -20
Command 'lsinitramfs' not found, but can be installed with:
sudo apt install initramfs-tools-core
alantang@LAPTOP-CV1Q6677:~$ cat /proc/cmdline
initrd=linuxrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug pty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
alantang@LAPTOP-CV1Q6677:~$ nano bootloader_concept.asm
```

10. Operating-System Debugging

10.1: Using strace

- `strace ls` # Trace system calls
- `strace -v ls` # Trace dengan detail
- `strace -c ls` # Hitung waktu system calls
- `strace -o trace.txt ls` # Trace dan simpan ke file
- `cat trace.txt`

10.2: using ltrace

- `ltrace ls` # Trace library calls
- `ltrace -c ls` # Dengan detail

10.3: Using GDB

- buat program buggy.c

Debug dengan GDB

- `gcc -g -o buggy buggy.c` # Compile dengan debug symbols
- `gdb ./buggy` # Jalankan dengan GDB

10.4: Kernel Debugging dengan dmesg

- `dmesg` # Lihat kernel log

```
slanganteng@LAPTOP-CV1QG677:~$ strace ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
slanganteng@LAPTOP-CV1QG677:~$ strace -v ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
slanganteng@LAPTOP-CV1QG677:~$ strace -c ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
slanganteng@LAPTOP-CV1QG677:~$ strace -o trace.txt ls
Command 'strace' not found, but can be installed with:
sudo apt install strace
slanganteng@LAPTOP-CV1QG677:~$ cat trace.txt
cat: trace.txt: No such file or directory
slanganteng@LAPTOP-CV1QG677:~$ ltrace ls
Command 'ltrace' not found, but can be installed with:
sudo apt install ltrace
slanganteng@LAPTOP-CV1QG677:~$ ltrace -c ls
Command 'ltrace' not found, but can be installed with:
sudo apt install ltrace
slanganteng@LAPTOP-CV1QG677:~$ nano buggy.c
slanganteng@LAPTOP-CV1QG677:~$ gcc -g -o buggy buggy.c
slanganteng@LAPTOP-CV1QG677:~$ gdb ./buggy
Command 'gdb' not found, but can be installed with:
sudo apt install gdb
slanganteng@LAPTOP-CV1QG677:~$ dmesg
[ 0.000000] Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439a258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun 5 18:30:46
UTC 2025
[ 0.000000] Command line: initrd=initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count=0 WSL_ENABLE_CRASH_DUMP=1
[ 0.000000] KERNEL supported cpus:
[ 0.000000] Intel GenuineIntel
[ 0.000000] AMD AuthenticAMD
[ 0.000000] BIOS-provided physical RAM map:
[ 0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] usable
[ 0.000000] BIOS-e820: [mem 0x00000000000e0000-0x00000000000e0fff] reserved
[ 0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000001fffff] ACPI data
[ 0.000000] BIOS-e820: [mem 0x0000000000200000-0x00000000007fffff] usable
[ 0.000000] BIOS-e820: [mem 0x0000000010000000-0x000000001f2bffff] usable
[ 0.000000] NX (Execute Disable) protection: active
[ 0.000000] APIC: Static calls initialized
[ 0.000000] DMU not present or invalid.
[ 0.000000] Hypervisor detected: Microsoft Hyper-V
[ 0.000000] Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0xc2c, misc 0xe0bed7b6
[ 0.000000] Hyper-V: Nested features: 0x4a0000
[ 0.000000] Hyper-V: LAPIC Timer Frequency: 0xc3500
```


- dmesg -w # Follow kernel log

```
alangan@LAPTOP-CV1Q6777:~$ dmesg -w
0.000000 Linux version 6.6.87.2-microsoft-standard-WSL2 (root@439e258ad544) (gcc (GCC) 11.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP PREEMPT_DYNAMIC Thu Jun  5 18:30:46
UTC 2025
0.000000 Command line: initrd=\initrd.img WSL_ROOT_INIT=1 panic=1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hvc0 debug ptty.legacy_count=0 WSL_ENABLE_CRASH_DUMP
=1
0.000000 KERNEL supported cpus:
0.000000 Intel GenuineIntel
0.000000 AMD AuthenticAMD
0.000000 BIOS-provided physical RAM map:
0.000000 BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] usable
0.000000 BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] reserved
0.000000 BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] ACPI data
0.000000 BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] usable
0.000000 BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] usable
0.000000 BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] usable
0.000000 NX (Execute Disable) protection: active
0.000000 APIC: Static calls initialized
0.000000 DMI not present or invalid.
0.000000 Hypervisor detected: Microsoft Hyper-V
0.000000 Hyper-V: privilege flags low 0xae7f, high 0x3b8030, hints 0xc2c, misc 0xe0bed7b6
0.000000 Hyper-V: Nested Features: 0x400000
0.000000 Hyper-V: LAPIC Timer Frequency: 0xc3500
0.000000 Hyper-V: Using hypercall for remote TLB flush
0.000000 clocksource: hyperv_clocksource_tsc_page: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
0.000000 clocksource: hyperv_clocksource_msr: mask: 0xffffffffffffff max_cycles: 0x24e6a1710, max_idle_ns: 440795202120 ns
0.000000 tsc: Detected 1796.563 Mhz processor
0.000122 e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
0.000160 e820: remove [mem 0x00000000-0x00000fff] usable
0.000168 last_pfn = 0x1f2c00 max_arch_pfn = 0x400000000
0.000219 MTRR map: 5 entries (4 fixed + 1 variable; max 20), built from 8 variable MTRRs
0.000225 x86 PAT: Configuration [0-7]: WB WC UC- UC WB WP UC- WT
0.000316 last_pfn = 0xf5000 max_arch_pfn = 0x400000000
0.000342 Using GB pages for direct mapping
0.000760 RAMDISK: [mem 0x04be4000-0x04e7ffff]
0.000766 ACPI: Early table checksum verification disabled
0.000772 ACPI: RSDP 0x0000000000000000 000024 (v02 VIRTUAL)
0.000780 ACPI: XSDT 0x0000000000000000 000044 (v01 VIRTUAL MICROSFT 00000001 MSFT 00000001)
0.000790 ACPI: FACP 0x0000000000010100 000114 (v06 VIRTUAL MICROSFT 00000001 MSFT 00000001)
0.000797 ACPI: DSDT 0x0000000000010118 01E191 (v02 MSFTVM DSDT01 00000001 MSFT 05000000)
0.000803 ACPI: FACS 0x0000000000010114 000040
0.000807 ACPI: OEM0 0x0000000000010114 000004 (v01 VIRTUAL MICROSFT 00000001 MSFT 00000001)
0.000812 ACPI: SRAT 0x000000000001F349 000380 (v02 VIRTUAL MICROSFT 00000001 MSFT 00000001)
0.000817 ACPI: APIC 0x000000000001F6F9 0000C8 (v04 VIRTUAL MICROSFT 00000001 MSFT 00000001)
```

- dmesg -l err # Filter by level
- dmesg -l warn

10.5: Memory Leak Detection dengan Valgrind

- sudo apt install valgrind # Install valgrind

```
alangan@LAPTOP-CV1Q6777:~$ dmesg -l err
0.312444 PCI: Fatal: No config space access function found
3.604289 misc dxg: dxgk: dxgkio_is_feature_enabled: Ioctl failed: -22
3.614855 misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
3.616588 misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
3.618294 misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -22
3.620086 misc dxg: dxgk: dxgkio_query_adapter_info: Ioctl failed: -2
11.739644 WSL (274) ERROR: CheckConnection: getaddrinfo() failed: -3
16.746510 WSL (274) ERROR: CheckConnection: getaddrinfo() failed: -3
70.643494 WSL (274) ERROR: CheckConnection: connect() failed: 101

alangan@LAPTOP-CV1Q6777:~$ dmesg -l warn
0.216485 Speculative Return Stack Overflow: IBPB-extending microcode not applied!
0.216486 Speculative Return Stack Overflow: WARNING: See https://kernel.org/doc/html/latest/admin-guide/hw-vuln/srso.html for mitigation options.
0.373384 PCI: System does not support PCI
0.788159 device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate IMA measurements will not be recorded in the IMA log.
3.348337 pulseaudio[305]: memfd_create() called without MFD_EXEC or MFD_NOEXEC_SEAL set
3.708742 Failed to connect to bus: No such file or directory
4.250823 systemd-journal[46]: File /var/log/journal/263dfe39f1b741789eee8dfb5eed78/system.journal corrupted or uncleanly shut down, renaming and replacing.
883.832108 TCP: eth0: Driver has suspect GRO implementation, TCP performance may be compromised.

alangan@LAPTOP-CV1Q6777:~$ sudo apt install valgrind
[sudo] password for alangan:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm9
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libipt2 libsource-highlight-common libsource-highlight4t64
Suggested packages:
  gdb-doc gdbserver valgrind-dbg valgrind-mpi kcachegrind alleyoop valkyrie
The following NEW packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libipt2 libsource-highlight-common libsource-highlight4t64 valgrind
0 upgraded, 9 newly installed, 0 to remove and 0 not upgraded.
Need to get 26.9 MB of archives.
After this operation, 105 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod-common all 0.190-1.1ubuntu0.1 [14.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginfod1t64 amd64 0.190-1.1ubuntu0.1
```

- cat > memleak.c << 'EOF' # Buat program dengan memory leak
- gcc -g -o memleak memleak.c # Compile
- valgrind --leak-check=full ./memleak # Jalankan dengan valgrind

10.6: Performance Analysis

- sudo apt install linux-tools-common linux-tools-generic # CPU profiling dengan perf

```
alanganteng@LAPTOP-CV1QG677:~$ cat > memleak.c << 'EOF'
> #include <stdlib.h>
> int main() {
>     int *ptr = (int*)malloc(100 * sizeof(int));
>     // Forgot to free(ptr)
>     return 0;
> }
> EOF
alanganteng@LAPTOP-CV1QG677:~$ gcc -g -o memleak memleak.c
alanganteng@LAPTOP-CV1QG677:~$ valgrind --leak-check=full ./memleak
==4088== Memcheck, a memory error detector
==4088== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==4088== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==4088== Command: ./memleak
==4088==
==4088== HEAP SUMMARY:
==4088==    in use at exit: 400 bytes in 1 blocks
==4088== total heap usage: 1 allocs, 0 frees, 400 bytes allocated
==4088==
==4088== 400 bytes in 1 blocks are definitely lost in loss record 1 of 1
==4088==    at 0x4846828: malloc (in /usr/libexec/valgrind/vgpreload_memcheck-amd64-linux.so)
==4088==    by 0x10915E: main (memleak.c:3)
==4088==
==4088== LEAK SUMMARY:
==4088==    definitely lost: 400 bytes in 1 blocks
==4088==    indirectly lost: 0 bytes in 0 blocks
==4088==    possibly lost: 0 bytes in 0 blocks
==4088==    still reachable: 0 bytes in 0 blocks
==4088==    suppressed: 0 bytes in 0 blocks
==4088==
==4088== For lists of detected and suppressed errors, rerun with: -s
==4088== ERROR SUMMARY: 1 errors from 1 contexts (suppressed: 0 from 0)
alanganteng@LAPTOP-CV1QG677:~$ sudo apt install linux-tools-common linux-tools-generic
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  liblvm2
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  hwdata libnuma1 libpci3 libtraceevent1 linux-tools-6.8.0-85 linux-tools-6.8.0-85-generic
The following NEW packages will be installed:
```

- sudo perf trace ls # Monitor system calls
- sudo perf top # Top-like interface

```
alanganteng@LAPTOP-CV1QG677:~$ sudo perf trace ls
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
alanganteng@LAPTOP-CV1QG677:~$ sudo perf top
WARNING: perf not found for kernel 6.6.87.2-microsoft

You may need to install the following packages for this specific kernel:
  linux-tools-6.6.87.2-microsoft-standard-WSL2
  linux-cloud-tools-6.6.87.2-microsoft-standard-WSL2

You may also want to install one of the following packages to keep up to date:
  linux-tools-standard-WSL2
  linux-cloud-tools-standard-WSL2
```